BIOL 230 MIDTERM 1 REVIEW SHEET

1. Introduction to Ecology

A. Core Concepts:

- Ecology = scientific study of interactions between organisms and their environment (including other organisms)
- Adaptation = heritable trait that improves survival/reproduction in a given environment (Concept 1.3).
- Natural selection = individuals with advantageous traits survive & reproduce more; frequencies of those traits increase over time.
- Producer (autotroph) = captures external energy (e.g., sunlight) to make food.
- Consumer (heterotroph) = gains energy by eating organisms/remains.
- Net Primary Production (NPP) = energy captured by producers – energy lost as respiration.
- Nutrient cycle = movement of materials (C, N, P, etc.) between organisms & the environment.
- Competitive release = species increases when competitor is removed (e.g., Evasterias after Pisaster declined).

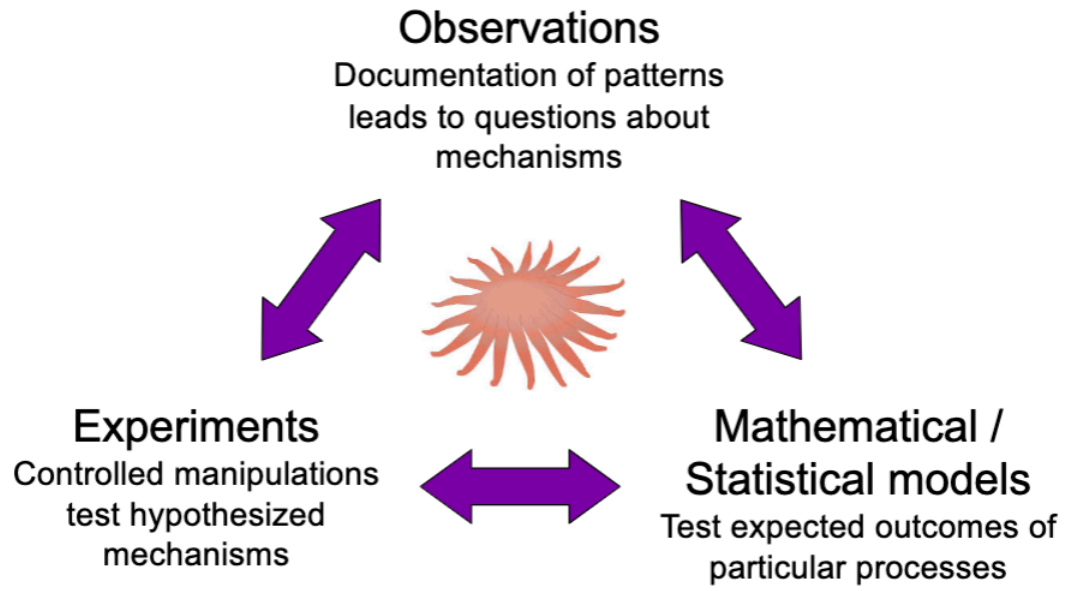
B. How Ecologists Ask Questions (Concept 1.3)

Ecologists evaluate competing hypotheses using:

- Observations – document natural patterns.
    - e.g., Frogs with deformities only in ponds with snails carrying Ribeiroia.
- Experiments – manipulate variables, include controls & replication.
    - Lab experiment confirmed *Ribeiroia* caused deformities.
- Models – mathematical/statistical, test expected outcomes.

Field vs Lab Experiments

- *Field*: Realistic but less control.
- *Lab*: High control, less ecological realism.
- *Models*: Flexible, but depend on assumptions.



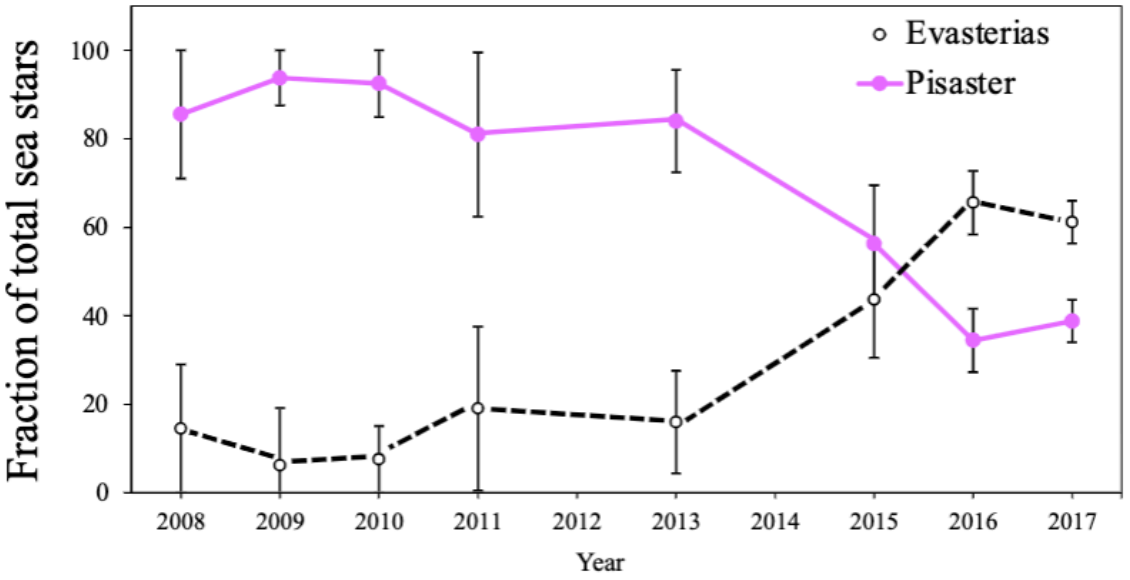
C. Case Studies from Lecture 02

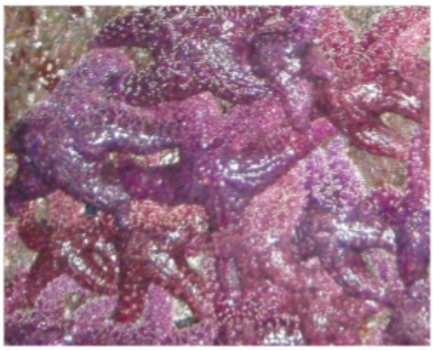
① Shipworms & the Spanish Armada

- Pattern: Spanish ships decayed more than English ones.
- Mechanism: Warm waters → higher shipworm activity.
- Big idea: Organisms (shipworms) can alter human history via ecological interactions.

② Sea Star Wasting Disease (SSWD)

- Observation: Pisaster populations collapsed, Evasterias increased.





Pisaster

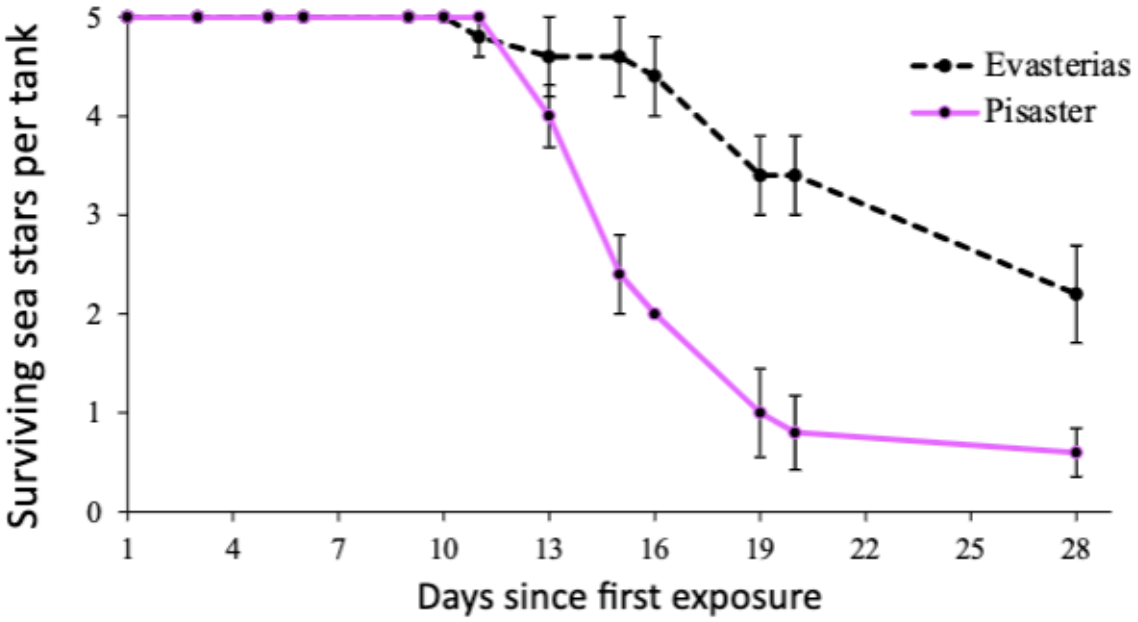


Evasterias

- Hypotheses:
    - Evasterias less susceptible to disease.
    - Competitive release:

      (a) Exploitation (shared prey → more food).

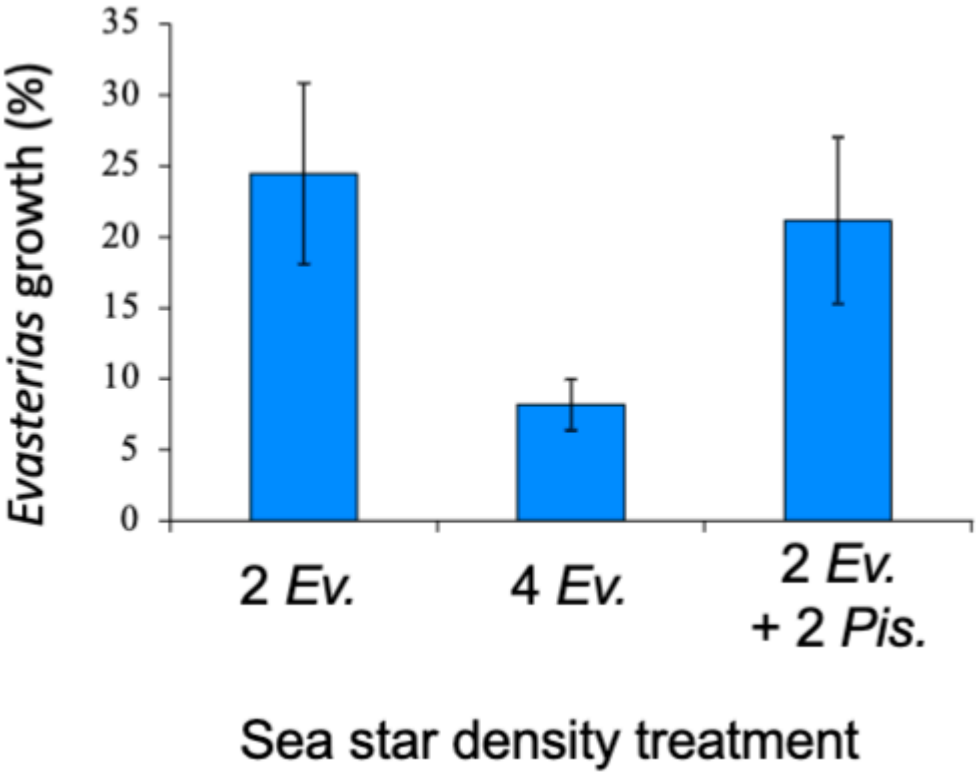
      (b) Interference (Pisaster bullies Evasterias).
- Experiments:
    - Susceptibility test → Evasterias survived longer.✔



- Diet test → diets similar. ✔

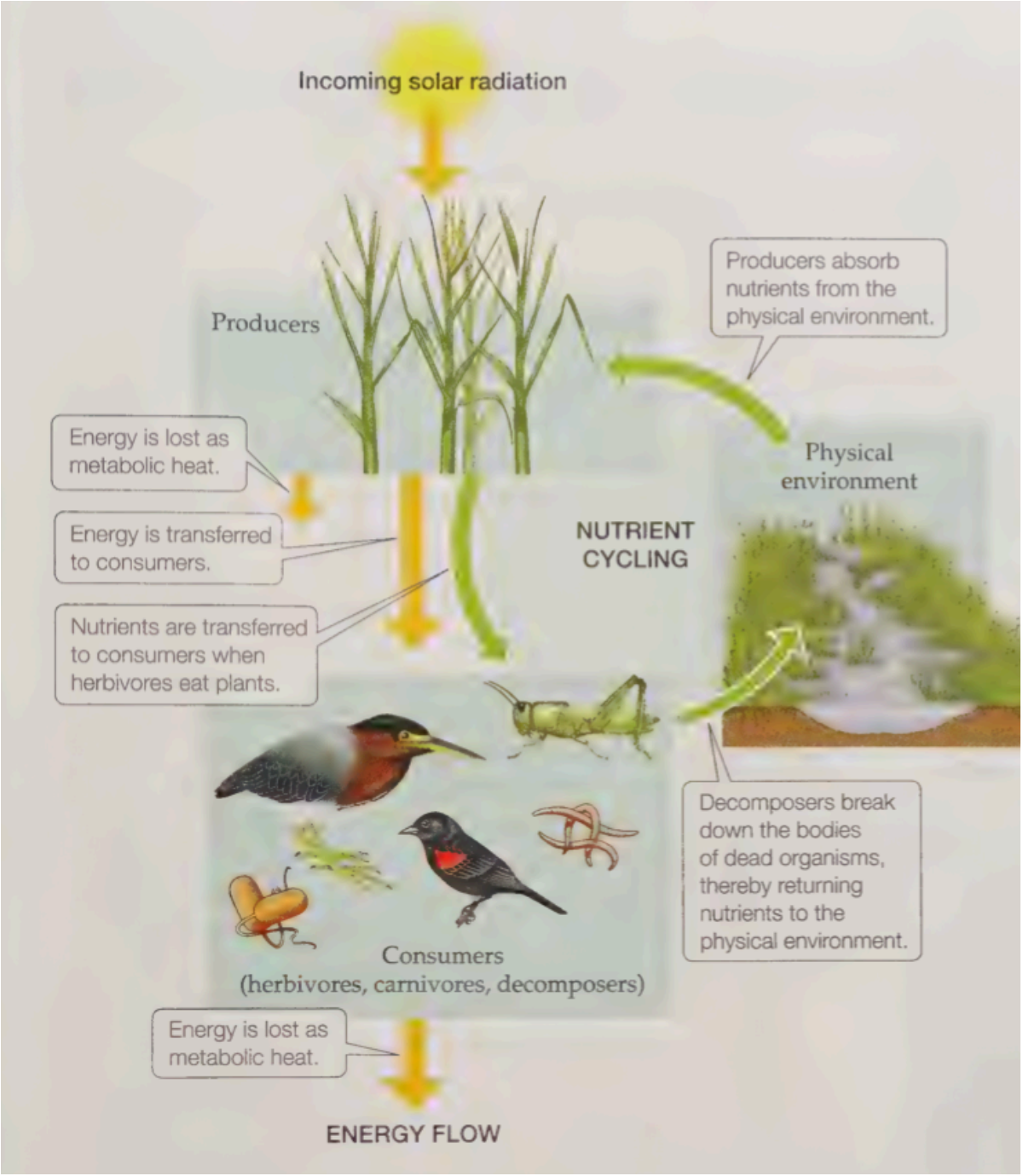
predator	prey	per cent of meals (*n*)
Evasterias	barnacles	47.4 (9)
	mussels	57.9 (11)
Pisaster	barnacles	39.5 (15)
	mussels	73.7 (28)

- Competition test → Pisaster does not affect Evasterias growth. ✗



D. Energy & Nutrients

- Energy: One-way flow → always lost as heat.
- Nutrients: Recycled (e.g., phosphorus, nitrogen).



2. Introduction to Ecology continued

A. Core Definitions:

- Ecology = the scientific study of interactions between organisms and their environment (including other organisms)

- Alternate definition = study of interactions that determine **distribution and abundance** of organisms

👉 Key testable point: Be able to state both definitions and explain their difference in emphasis.

B. Ecology vs Environmental Science

- Ecology: biology discipline, focused on mechanisms and processes in nature.
- Environmental science: interdisciplinary (includes politics, economics, ethics), focused on solving human-driven environmental problems.

👉 Key testable point: Distinguish clearly between ecology (scientific) and environmental science (applied + interdisciplinary).

C. Spatial & Temporal Scales

- Spatial scale:
  - Individual → population → community → ecosystem → landscape → biosphere
  - Each level nests within the one above.
- Temporal scale:
  - Seconds → years/decades → millennia.
  - Different processes act at different time scales (e.g., predation vs climate change).

👉 Key testable point: Be able to order levels of organization and give examples at different spatial/temporal scales.

D . Case Study: Western Hemlock & Looper Moth

- Distribution/abundance of western hemlock influenced by:
  - Abiotic: temperature, precipitation, light, soil, elevation.
  - Biotic: herbivores (e.g., hemlock looper), pathogens, mutualists.

- Looper moth outbreaks:
    - Periodic, climate-linked defoliation events.
    - Not uniform → context matters (local environment affects outcome).

☐ What conclusions can we draw regarding potential outbreak triggers from these data?

- Outbreaks occur in years with warmer than average Junes and sometimes lower precipitation.
- Climate variation seems to play a role in triggering outbreaks.

☐ Why don't outbreaks occur everywhere every time? Why do some sites only outbreak once and others outbreak regularly?

- Local abiotic conditions (temperature, moisture, soil) and biotic context (presence of natural enemies, host density, forest structure) differ between sites.

👉 Key testable point: Factors controlling species distribution = abiotic + biotic + interactions with climate.

E. Predictability in Ecology

- **Physics vs Ecology**:
    - Physics → universal laws, highly predictable.
    - Ecology → patterns exist but are **context-dependent, noisy, stochastic**
- Some predictability at **large scales** (e.g., biogeography), but less so at community level.

☐ Can we answer ecological questions with certainty?

- Not always. Ecological systems are complex and noisy, with many interacting variables and context dependence.
- Some patterns are repeatable, but universal, law-like predictability is rare.

👉 **Key testable point**: Explain why ecology is less "law-like" than physics.

F. Five Key Processes Shaping Distribution & Abundance

1. **Abiotic requirements** – temperature, water, nutrients.
2. **Biotic interactions** – competition, predation, mutualism, pathogens.
3. **Dispersal** – ability to reach suitable habitats.
4. **Evolution** – adaptation of traits to environment.
5. **Stochasticity** – randomness (disturbances, chance events).

👉 **Key testable point**: List and define all five processes.

3. Welcome to Earth

A. Key Concepts

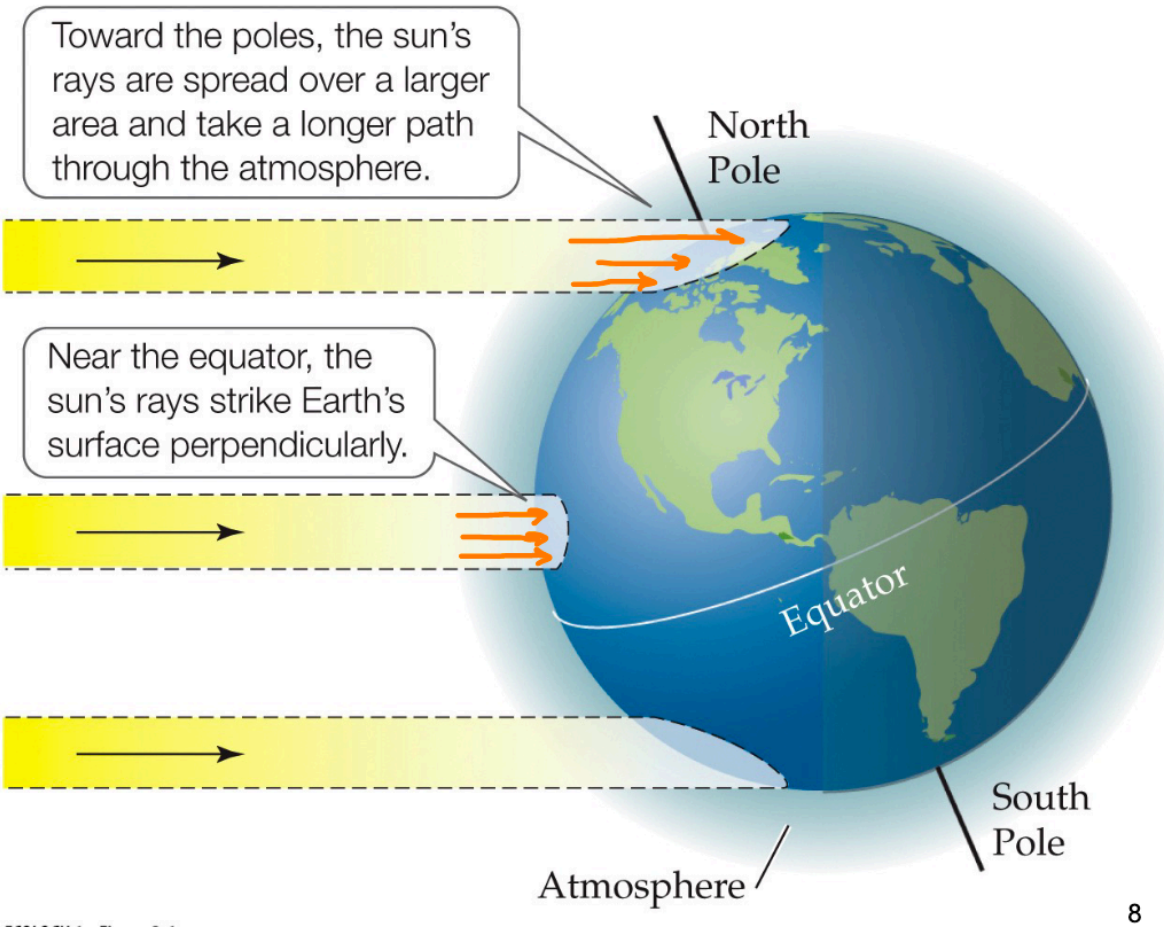
- Climate is the most fundamental environmental factor.
- Winds & ocean currents arise from unequal solar heating.
- Circulation patterns drive global temperature & precipitation.
- Regional climates shaped by continents, oceans, mountains, vegetation.

B. Climate: the environmental backdrop

- Climate = long-term averages of temperature and precipitation; weather = short-term conditions.
- Climate is the most fundamental factor shaping ecology → it sets where organisms can live and how ecosystems function.
- Example: deserts vs rainforests differ mainly because of climate, not because of the organisms themselves.

C. Why the tropics are warmer than the poles

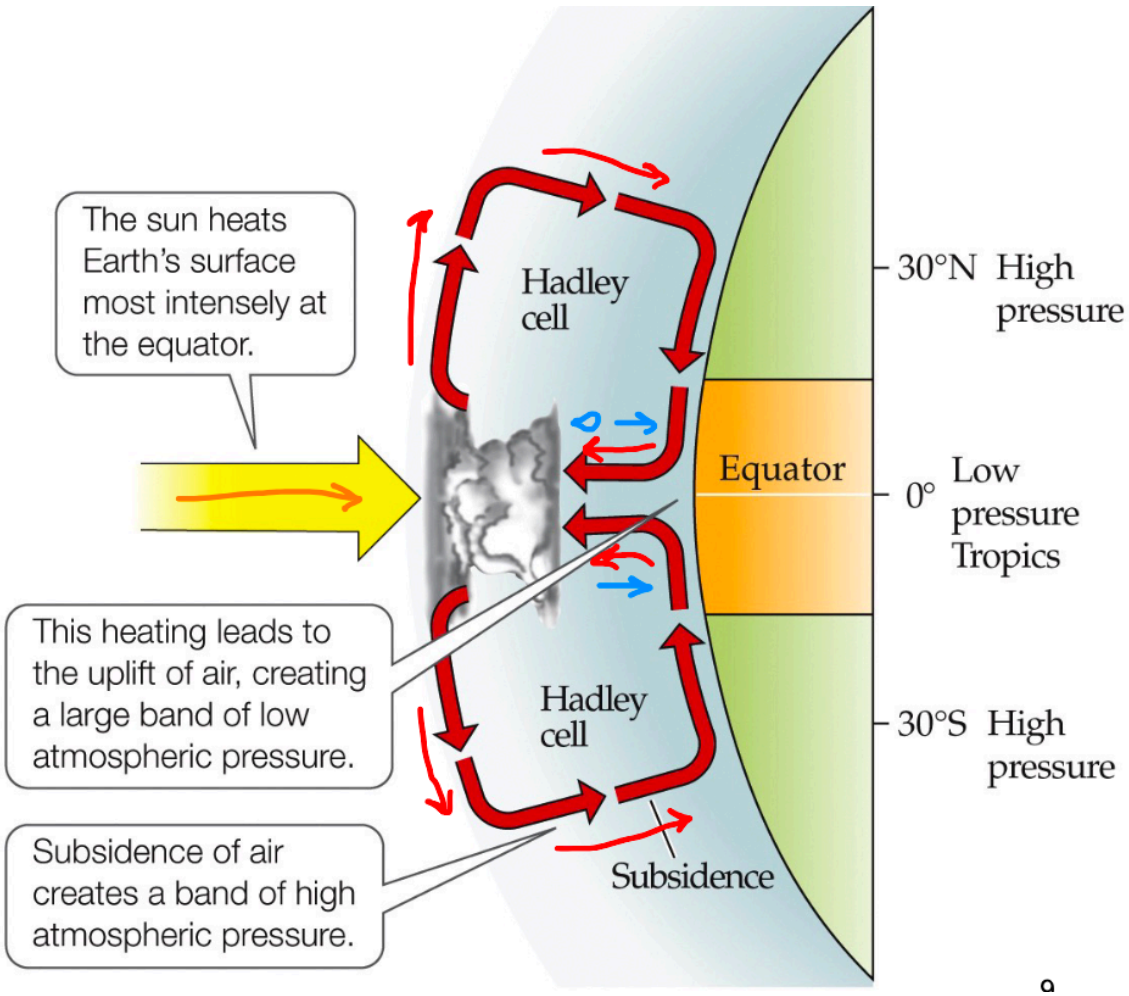
- Uneven solar radiation:
    - At the equator, sunlight hits directly → energy is concentrated.
    - At the poles, sunlight arrives at an angle → energy is spread out and passes through more atmosphere.



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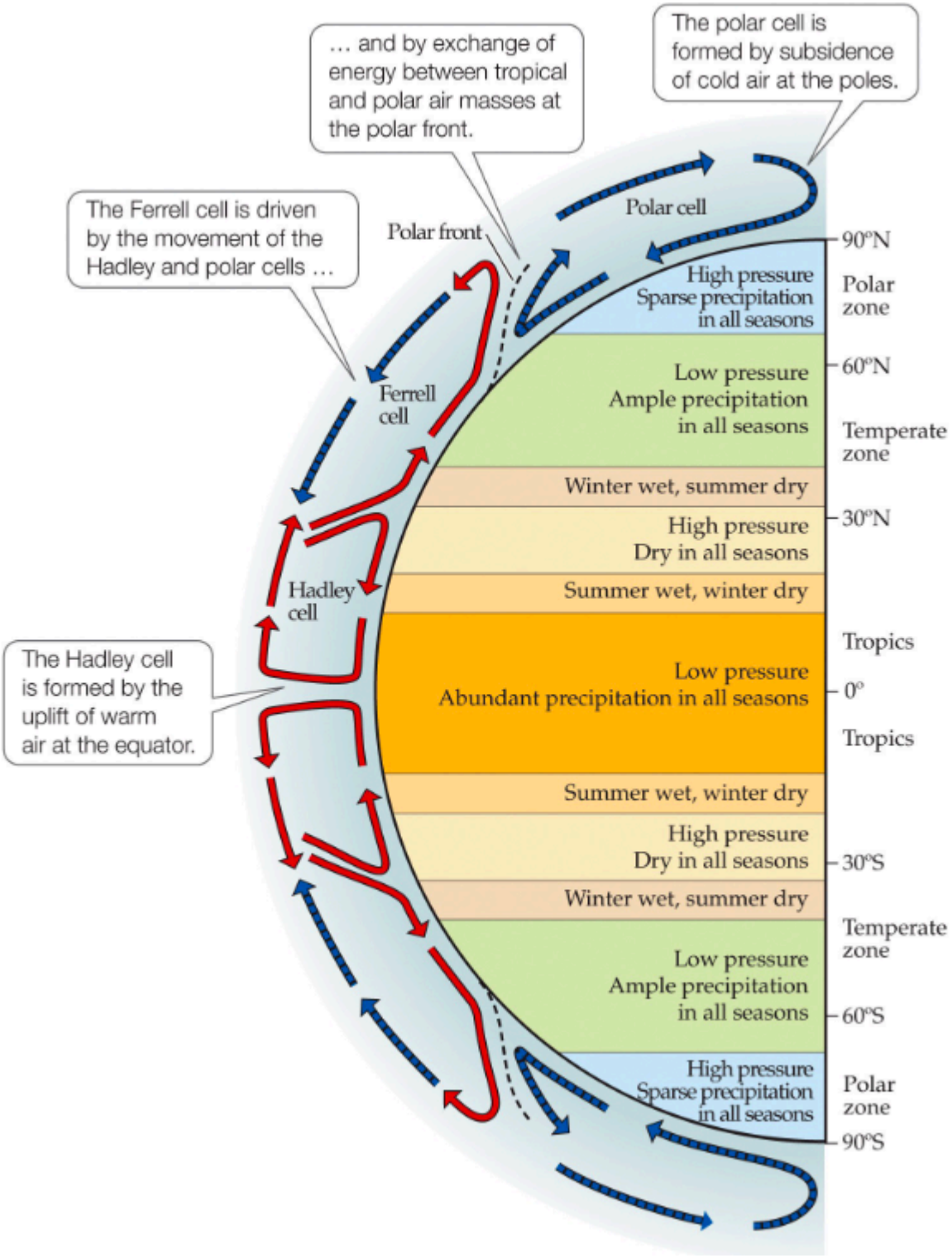
- Air movement: warm air rises at the equator → as it rises it cools → cooler air holds less moisture → rain falls → explains tropical rainfall.
- After losing moisture, dry air sinks at ~30° N/S → warms as it descends → creates deserts.

  👉 This system of rising at the equator and sinking at 30° is the Hadley cell.



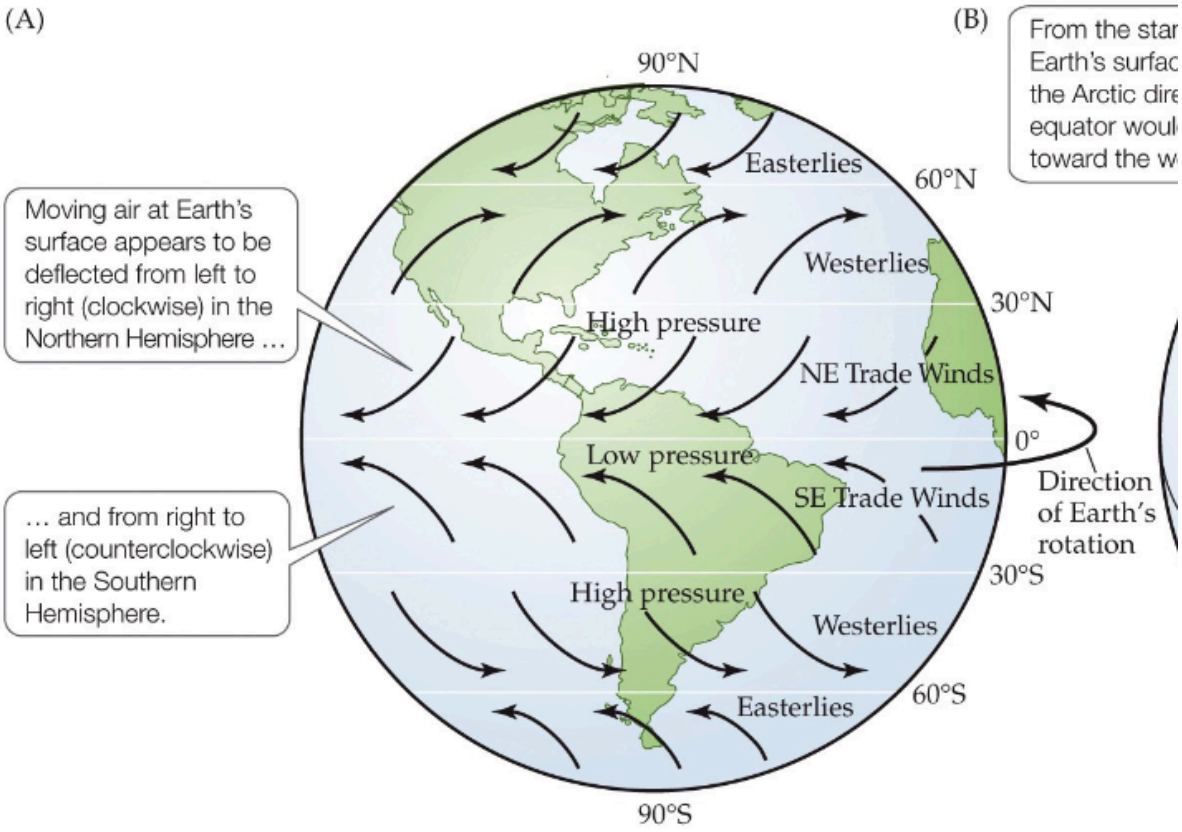
D. Winds, circulation & precipitation

- The 3-cell model:
    - Hadley cell (0–30°) → wet tropics, dry subtropical deserts.
    - Ferrel cell (30–60°) → mixing zone, variable climate.
    - Polar cell (60–90°) → cold, dry air sinks at poles.



- Coriolis effect: Earth's rotation deflects moving air:
    - Rightward in Northern Hemisphere, leftward in Southern Hemisphere.
    - This deflection turns N–S winds into trade winds, westerlies, polar easterlies.

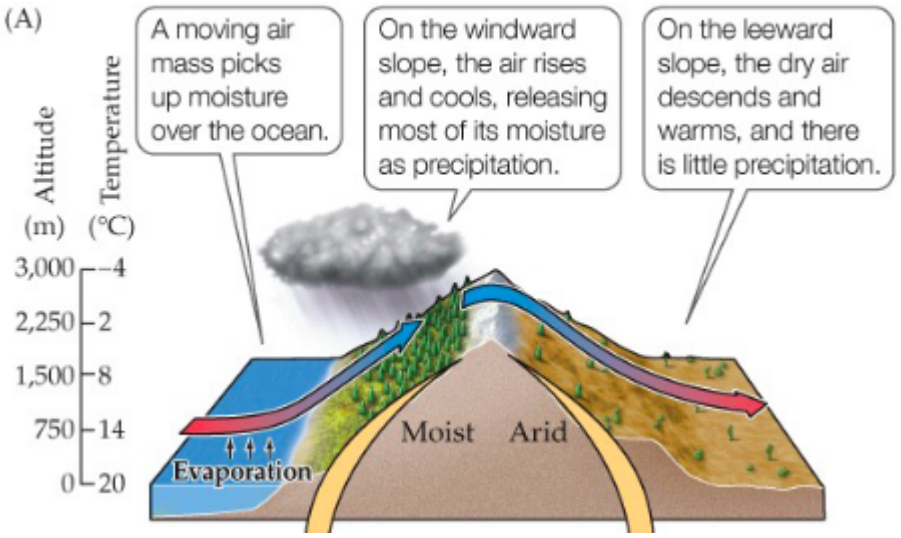
- Winds + rotation drive ocean currents, which move heat and nutrients around the globe.



E. Regional climate influences

Even though global circulation explains broad patterns, local factors create variation:

1. Mountains → Rain shadow effect
    - Windward side: moist air rises → cools → heavy rainfall.
    - Leeward side: dry descending air → creates arid conditions.
    - Example: Andes → Amazon wet on east, Atacama desert on west.



2. Oceans
    - Store heat, so coastal areas are wetter and milder (cooler summers, warmer winters).
    - Example: Vancouver Island wetter and warmer in winter than PEI.
3. Continents
    - Interiors far from oceans heat and cool more dramatically → stronger seasonal extremes.
4. Vegetation
    - Forests vs grasslands change **albedo** (reflectivity) and evapotranspiration, feeding back on climate.
5. Elevation
    - Higher altitude = cooler temperatures,wetter, different precipitation and vegetation.
    - BC examples: Cypress Bowl, Hope Valley, Manning Park all show distinct plant communities by elevation and climate.

F. Case Studies / Examples

1. Why is Mexico warmer than Canada?

- Latitude: Mexico receives more direct solar radiation.

2. Andes Mountains
    - Example of rain shadow deserts on leeward slopes.
3. BC sites (Cypress Bowl, Hope Valley, Manning Park, Princeton, Kelowna)
    - Show how temperature, precipitation, and elevation shape plant communities.

G. Discussion / Clicker Questions

- Q: What does warm air rising and cooling mean for precipitation?

  → High rainfall at equator, low rainfall at ~30°.
- Q: Where would you expect rain shadows?

  → Always on the leeward side of mountains.
- Q: Why is Vancouver Island wetter than PEI?

  → Pacific Ocean moisture + mountain uplift.
- Q: Why is Vancouver Island warmer in winter than PEI?

  → Ocean moderates seasonal extremes.
- Q: Flat Earth thought experiment

  → Without Earth's rotation & curvature: no Coriolis, no Hadley cells, no banded circulation/rainfall belts.

4. Coping with the environment

A. Key Concepts

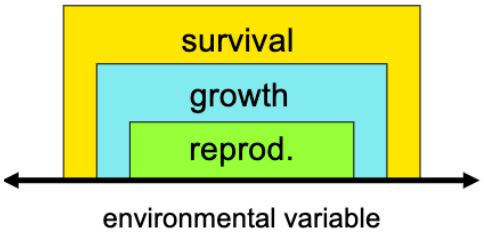
- 4.1 Each species has a tolerance range for abiotic factors; these tolerance windows set the potential geographic distribution.
- 4.2 An organism's body temperature is set by energy exchanges with the environment (gains vs losses).

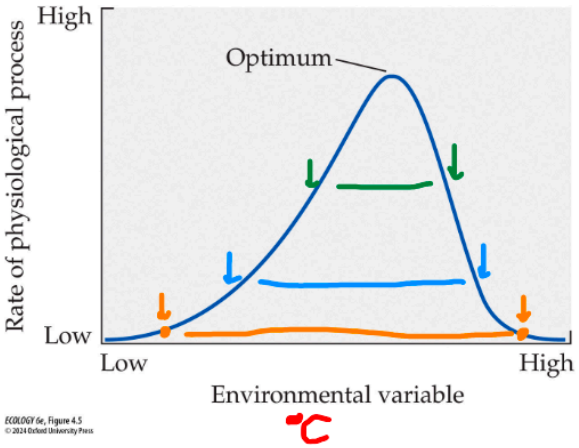
    - Body temperature controls physiology and survival → extreme limits ~ –5°C to ~50°C for most plants/animals.
    - **Temperature affects physiology** (enzyme function, membranes, water balance).
    - **Organisms cope** by tolerance, acclimatization, or active regulation
    - **Ectothermy vs endothermy**: two major strategies, each with costs/benefits for survival and distribution.
- What are some important abiotic (non-living) factors that affect organisms?
    - Temperature, light, soil quality, humidity, precipitation, oxygen, wind, water quality, salinity, nutrients

B. Environmental Temperature Variation

- Boreal forest: up to 80 °C swing between winter & summer.
- Tropics: only ~15 °C swing.
- Soils: buffered vs surface air temps.
- Aquatic systems:
    - Open ocean stable (huge heat capacity).
    - Tide pools fluctuate 20 °C within 5 hrs.
- Extreme cases: microbes in hot springs (90 °C), Antarctic fishes at –2 °C.
- Stress = reduced performance or cellular damage; always relative to organism.
    - –5 °C: stressful for tropical frog, not Arctic fox.

👉 Take-home: Different environments = different challenges → distributions reflect these limits.





C. Temperature and Physiology

- Enzymes: have optimal ranges.
    - Denature at 40–70 °C (proteins lose structure).
    - Stop functioning below –5 °C.
    - Hot-spring bacteria are stable up to 100 °C.
- Acclimation/Acclimatization:
    - Seasonal leaf pubescence changes;
    - intertidal organisms show plasticity after cold exposure.
    - Isozymes: multiple forms allow acclimatization (e.g., trout, carp, loblolly pine).
- Membranes:
    - Cold → lipids solidify, leak metabolites.
    - Cold-climate plants = more unsaturated lipids (flexible).
    - Tropical plants fail near 10 °C, alpine plants function ~0 °C
- Water coupling: warmer air holds more vapor → evaporative loss increases with

👉 Core idea: Temperature controls basic life chemistry. Too far out of range = death.

D. Energy Balance Basics

- Energy inputs:

    - Solar radiation (SR)
    - Infrared from environment (IR_in)
    - Conduction, convection (from warmer medium)
    - Metabolism (animals only)
- Energy outputs:
    - Infrared radiation (IR_out)
    - Conduction, convection (to cooler medium)
    - Evaporation / transpiration

Plants: $\Delta H_{\text{plant}} = SR + IR_{\text{in}} - IR_{\text{out}} + H_{\text{conv}} + H_{\text{cond}} - H_{\text{evap}}.$

Animals (add metabolic heat): $\Delta H_{\text{animal}} = SR + IR_{\text{in}} - IR_{\text{out}} + H_{\text{conv}} + H_{\text{cond}} - H_{\text{evap}} + H_{\text{met}}.$

- If inputs > outputs → body temp rises.
- If outputs > inputs → body temp drops.
- Signs: SR/IR_in warm; IR_out cools; convection/conduction warm or cool depending on gradient; **evapotranspiration always cools**.

E. Plant vs animal strategies (high yield)

- **Plants:**
    - Transpiration (stomata) → evaporative cooling.
    - Pubescence (leaf hairs) → reflect sunlight, but reduce convective cooling.
    - Leaf traits: size, orientation, roughness affect boundary layer and heat loss.
    - Examples: Encelia farinosa uses pubescence; alpine snow lotus use thick pubescence traps heat for emergent flowering stems.
- **Animals:**
    - **Evaporation: sweating, panting, licking.**
    - **Insulation: fur, feathers, fat (seasonally variable).**

- **Behavior: bask/shade, burrow, posture, microhabitat choice.**
- **Metabolic heat:**
  - **birds/mammals**
  - tuna swimming muscle generates heat
  - bees' heat balls

F. Ectotherms vs Endotherms

	Ectotherms	Endotherms
Heat source	External exchange	Internal H_{met}
Body-T tolerance	Broad; activity constrained by ambient T	Narrow; activity possible across climates
Size constraint	Large ectotherms rare (SA:V & exchange limits)	Small endotherms lose heat fast → high energy demand
Cold solutions	Avoid (migrate/burrow/microhabitat) or tolerate (cryoprotectants; rare freeze tolerance)	**TNZ** & **LCT**; below LCT metabolic heat rises; **torpor/hibernation** for energy savings

- "TNZ & LCT" refers to the Thermoneutral Zone (TNZ), a range of ambient temperatures at which an animal can maintain its core body temperature without an increase in metabolic rate, and its Lower Critical Temperature (LCT), the lowest temperature within the TNZ, below which the animal must expend metabolic energy to generate heat to stay warm.

G. Discussion Questions:

1. What is **stress**?
   - **Definitions: (1) reduced physiological rate (ecology), (2) cellular damage (physiology). Always relative to organism**

2. Coping strategies: Two approaches → Avoidance vs Tolerance (e.g., shade-seeking vs physiological tolerance)

Case Study:

1. Mussels (Mytilus californianus):

- Lab acclimation (10 weeks in cool, warm, hot conditions) shifted survival when exposed to extreme T. Field acclimatization + long-term adaptation also seen. Shows plasticity in thermal tolerance

2. Winter avoidance:

- Intertidal organisms in sheltered sites stayed warmer than those exposed, demonstrating microhabitat choice as stress avoidance

3. Snails :

- After cold low-tide exposure, snails actively moved to warmer microhabitats, behaviorally adjusting T

5. Acquiring energy

A. Key Concepts

- 5.1 Energy sources: organisms acquire energy from sunlight, inorganic chemicals, or organic compounds (autotrophs vs heterotrophs).
- 5.2 Storage: autotrophs convert radiant/chemical energy into energy-rich C–C bonds (carbohydrates); these power heterotrophs via consumption.
- Core ideas from lecture: energy supply varies in space/time (e.g., light); organisms acclimatize to track that variation; trade-offs mean you can’t be perfect at everything.

B. Case Study — Toolmaking Crows (why energy matters)

- New Caledonian crows manufacture tools (hooked twigs; serrated Pandanus strips) to extract larvae/spiders from wood; tools are reused and increase foraging efficiency (lab crows even bent wire into hooks).
- Sophisticated, socially learned foraging improves energy intake per effort, paralleling human tool use.

C. Energy Pathways & Definitions (clean distinctions)

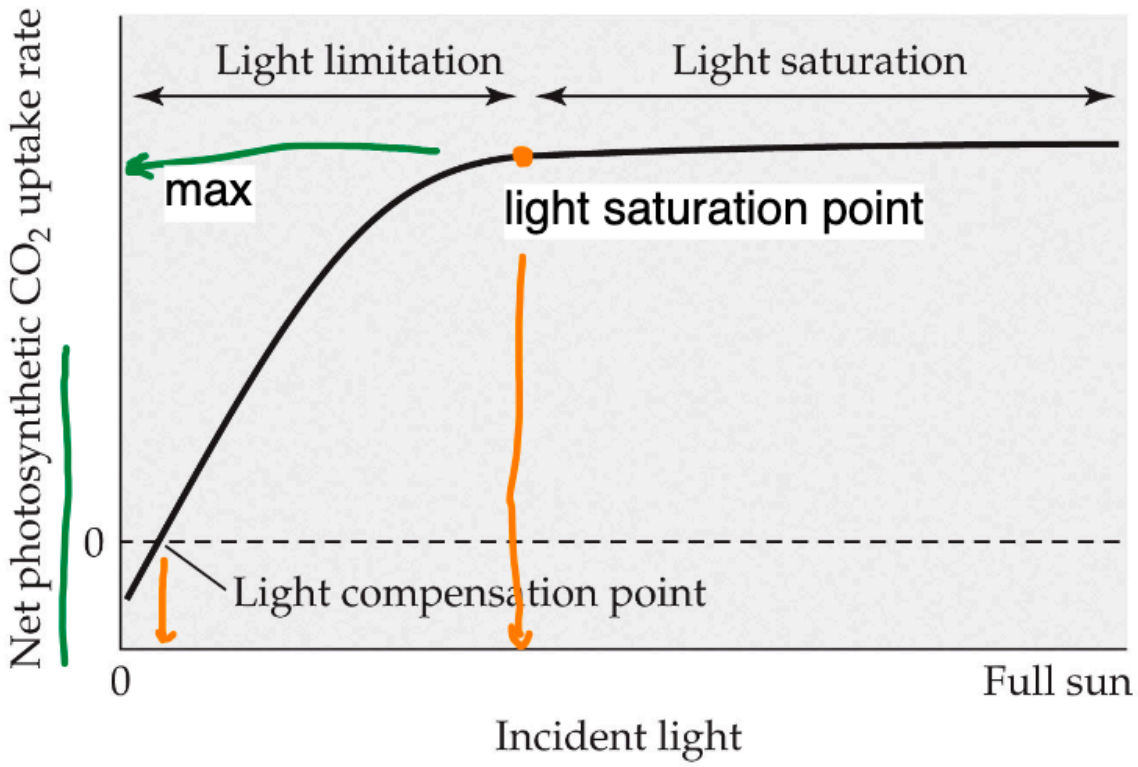
- Autotrophs
    - Photoautotrophs: capture sunlight → ATP & NADPH → fix CO_2 → carbohydrates (Calvin cycle; rubisco).
    - Chemoautotrophs: oxidize inorganic e^- donors → ATP/NADPH → fix CO_2.
        - Deep sea energy source? → **Chemosynthesis (oxidation of inorganics)**, plus marine snow.
- Heterotrophs: consume organic matter (predators, herbivores, parasites; detritivores).
- Edge cases:
    - Holoparasitic plants (e.g., dodder) are heterotrophs; hemiparasites (mistletoe) photosynthesize and tap hosts.
    - Some animals borrow photosynthesis (e.g., Elysia sea slugs keep algal chloroplasts) to supplement energy.

D. Photosynthesis — what to know

- Overall reaction: $6\ CO_2 + 6\ H_2O \rightarrow C_6H_{12}O_6 + 6\ O_2$.
- Two steps:
    - Light-driven reactions: pigments (chlorophylls + carotenoids) absorb photons → split H_2O, generate ATP/NADPH, release O_2.

- Carbon reactions (Calvin cycle): rubisco fixes CO_2 → 3-C intermediates → sugars (stored C–C energy).

- Light response curve: increases from light compensation point to light saturation (often below full sun). Plants acclimatize by shifting saturation/composition (leaf thickness, chloroplast density, pigment pools).



- **Special pigment**: chlorophyll f in cyanobacteria absorbs near-IR, enabling growth under/behind other photosynthesizers (low-light niches).
- **Water trade-off**: stomatal closure conserves water but limits CO_2 and raises risk of light damage (excess energy).
- **Temperature effects:** enzyme kinetics + membrane fluidity set **T-optima**; cold/alpine taxa photosynthesize near 0 °C; desert taxa peak ~40–50 °C; species can **acclimatize** by expressing different enzyme isoforms.

E. TRADE-OFFS :

- Mobile animals can flee stress but spend more energy.

- Introduced oysters tolerate higher T than native mussels, but can't reproduce in historically cooler BC waters.
- Acclimatization costs energy; helps if conditions worsen.
- Good at photosynthesis in cold or hot, not both.
- Low-light acclimation improves dim performance but reduces ability to capitalize on bright light.

6. Life histories

A. Key Concepts (7.1–7.4, integrated with lecture)

- **Life history** = the pattern of development, growth, reproduction, and survival for a species; the population-average pattern is its life history strategy.
- Strategies vary within & among species, shaped by genetics, environment, and plasticity; no strategy is perfect—trade-offs rule.

B. Core Terms

- **Allocation** (energy to growth vs reproduction vs survival); **trade-off** (improvement in one reduces another).
- **Asexual vs sexual** reproduction; **isogamy** (equal-sized gametes) vs **anisogamy** (eggs $\gg$ sperm).
- **Phenotypic plasticity** (one genotype → different phenotypes across environments).
- **Direct development** (no free-living larva) vs **indirect development** (larva → metamorphosis).
- **Paedomorphosis** (retain juvenile traits at maturity).
- Selection pressures differ by stage (egg/larva/juvenile/adult), so many species evolve complex life cycles (direct vs indirect development)

C. Life-history diversity & plasticity (how environment shapes traits)

- Within-species variation is ubiquitous (timing of maturity, clutch size, growth rate).

Plasticity examples:

- Ponderosa pine: cool/moist trees allocate more to leaves; desert trees shorter, thicker trunks (allocation shift).
- Spadefoot toad: omnivore vs carnivore tadpole morphs switch with diet; fast-growing carnivores favored in ephemeral ponds.
- Seasonal timing: climate change can break cue–environment match (e.g., snowshoe hare coat color mismatch → higher predation).

D. Reproduction modes & why they matter

- Asexual (binary fission, budding, clonal growth): fast, full-genome transmission; common in microbes, many plants/animals (e.g., corals clone to build colonies before spawning).
- Sexual: costs include "cost of males" (population growth halves) and recombination can break good gene combos; benefits = genetic variation, faster adaptation.
    - ☐ Pathogen test (case study): in C. elegans, pathogen exposure drives higher outcrossing and prevents extinction—sex helps in challenging environments.

☐ Nemo case: clownfish groups in one anemone are size-ranked; top fish = female, second = breeding male; if female dies, male changes sex. Shows sequential hermaphroditism, social control of reproduction, and dispersing larvae that later recruit to hosts.

E. Trade-offs you must recognize

- Offspring number vs size: with fixed energy, more young ⇒ smaller each (plants: seed size–number; lizards: northern pops = more, smaller eggs).
- Offspring number vs survival: beyond a threshold, adding offspring lowers survivorship (classic clutch-size logic).

- Current reproduction vs parent: investing now can reduce parental growth, survival, or future reproduction (trees: cone crops vs growth; animals: courtship effort vs lifespan).

F. Complex life cycles & stage-specific selection

- Why indirect development? Larvae can:
    - Use different habitats/resources, reduce intraspecific competition, and disperse widely (ocean currents).
- Why direct development?
    - Avoid risky larval habitats/predators; stay where survival is proven; avoid being "washed away."
- Examples:
    - Nassau grouper: juveniles shift from algal clumps to rocks at a size-dependent sweet spot (min predation, max growth).
    - Paedomorphic salamanders: can mature with gills and stay aquatic when that habitat is favorable (predation/food/competition dependent).
    - Nucella lamellosa: a direct-developing snail with crawl-away babies—limited dispersal makes populations sensitive to local stressors. TBT pollution (antifouling paint) induced reproductive failure ("imposex"), shrinking range despite adult hardiness.

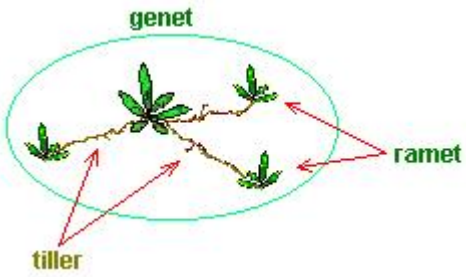
7. Population ecology I

A. Population & Individuals

- **Population**: Individuals of the same species living in the same area at the same time that interact (e.g., interbreeding, resource competition).
- **Abundance metrics:**

- **Population size** (N) vs. **Population density**: number of individuals vs. individuals per unit area (or volume).
- **Absolute vs relative size**: absolute counts/estimates(numbers) vs indices (tracks, CPUE, checklist tallies) that correlate with true abundance(ratio, percentage).

- Clonality:
  - **Genet**: A single genetic individual originating from one fertilization event.
  - **Ramet**: Physiologically independent unit of a clonally produced genet (may compete with other ramets).



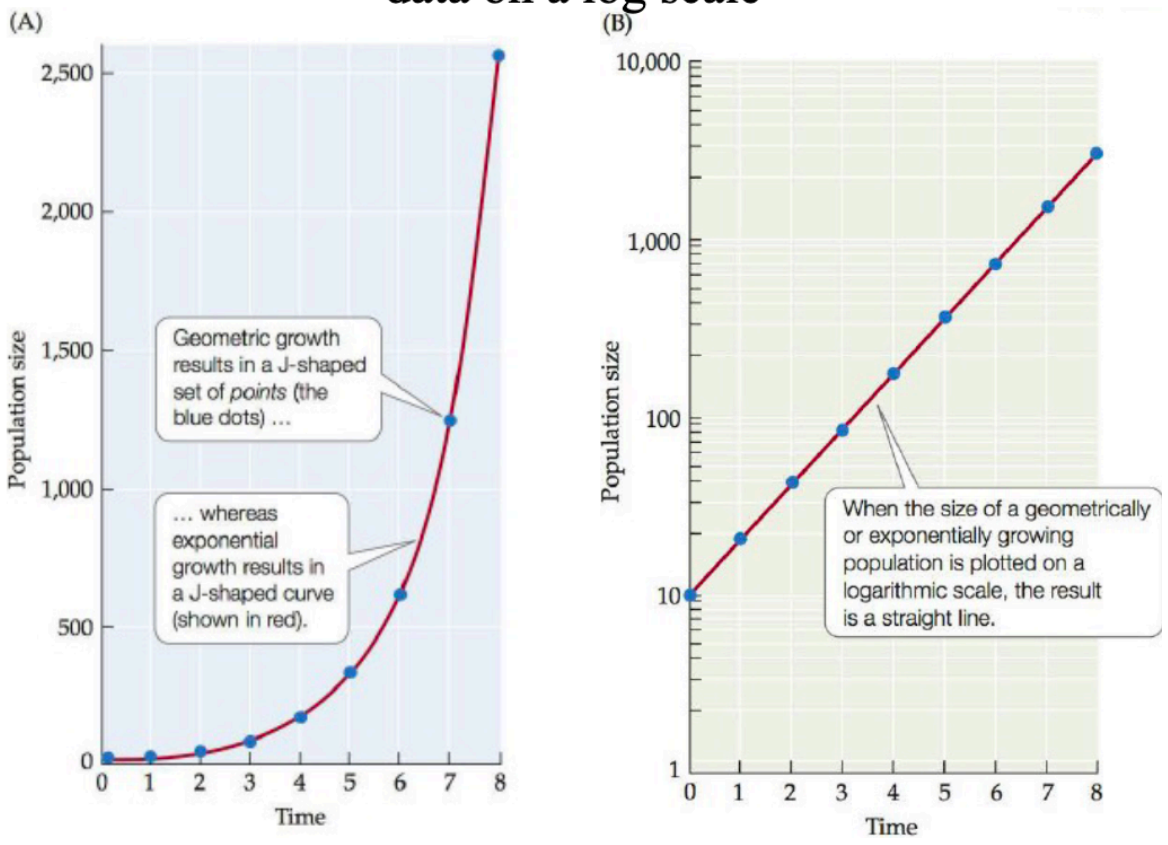
B. Growth models & equations

① Geometric growth (discrete time, synchronous reproduction) — use for organisms that reproduce at regular intervals (annual plants, periodical insects).

- N_t = population size after t generations or after t time periods (e.g. years)
- λ = geometric population growth rate
- N_0 = initial population size

- Recurrence: $N_{t+1} = \lambda N_t$
- Closed form: $N_t = N_0 \lambda^t$
- Interpretation:
  - $\lambda > 1 \rightarrow$ increase; $\lambda = 1 \rightarrow$ stable; $\lambda < 1 \rightarrow$ decline.
  - J-shaped points on linear scale; straight line on a log scale (log N vs. t).

Geometric/exponential curves become straight if you plot data on a log scale



$$\lambda = \frac{N_{t+1}}{N_t}$$

- Estimating λ from counts:

② Exponential growth (continuous time, overlapping reproduction) — use for organisms reproducing continuously (e.g., humans, microbes in batch culture).

- N(t)= the population size at each instant in time t
- N_0= initial population size
- dN/dt=rate of change in population size at each instant in time t

- r = exponential growth rate
- b = per-capita birth (natality) rate
    - Average births per individual per unit time. Units: 1/time (e.g., per year).
- d = per-capita death (mortality) rate
    - Average deaths per individual per unit time. Units: 1/time.

- Differential: $dN/dt = r N$ (where $r = b - d$ is the intrinsic rate of increase).
- Solution: $N(t) = N_0 e^{rt}$
- Interpretation:
    - $r > 0 \rightarrow$ increase; $r = 0 \rightarrow$ stable; $r < 0 \rightarrow$ decline.
    - J-shaped curve on linear scale; straight line on log scale (ln N vs. t).
- Estimating r over an interval:

$$r = \frac{\ln N_t - \ln N_0}{t} = \frac{\ln\left(\frac{N_t}{N_0}\right)}{t}$$

$$N_t = N_0 e^{rt} \Rightarrow N_t/N_0 = e^{rt} \Rightarrow \ln(N_t/N_0) = rt \Rightarrow r = \ln(N_t/N_0)/t.$$

③ Converting between λ and r

$$\lambda = e^r$$

$$r = \ln(\lambda)$$

-
- Use this to compare discrete and continuous formulations or match curves.

④ Doubling time

- t_d = time it takes a population growing exponentially to double in size.
- $t_d = \ln(2) / r$ (for exponential growth).

Exponential model: $N(t) = N_0 e^{rt}$.
"Double" means $N(t_d) = 2N_0$:

$$2N_0 = N_0 e^{rt_d} \Rightarrow 2 = e^{rt_d} \Rightarrow \ln 2 = rt_d \Rightarrow t_d = \frac{\ln 2}{r}.$$

Requires $r > 0$. If $r = 0$, $t_d = \infty$ (no growth). If $r < 0$, there's no doubling; use **halving time** $t_{1/2} = \frac{\ln 2}{|r|}$.

C. Measuring Abundance (choose method by organism & logistics)

1. Area-based counts (quadrats): count within plots, extrapolate to total area; best for sessile/slow organisms.
2. Distance methods (line/point transects): record detections and distances, fit a detection function to get density.
3. Mark–recapture (for mobile organisms): simplest estimate

$$\hat{N} \approx \frac{M_1 M_2}{R}$$

; key assumptions—closed population, equal catchability, marks retained/no behavior change.

M_1: number **Marked** in the **first** capture, then released.
M_2: total number **captured** in the **second** sample (marked + unmarked).
R: number in the second sample that are **Recaptures** (i.e., already marked).
$\hat{N}$: estimated total population size.

D. Case Studies

1. **Mice: Why populations can grow so fast — and why they don't forever**

- Growth is by multiplication, not addition; even modest r or λ yields rapid increases when N is large.
- Limits (resources, space, predators, disease, etc.) ensure exponential/geometric growth cannot persist indefinitely (numbers would otherwise become physically impossible).

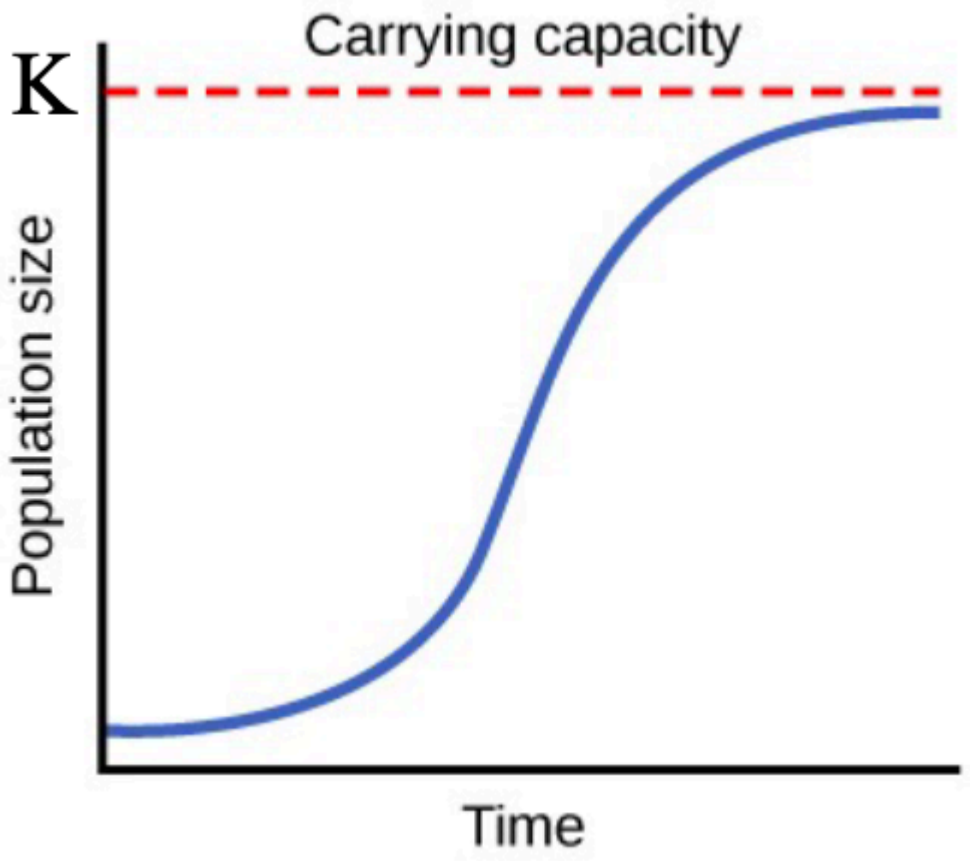
2. **Sockeye salmon: Distinct populations despite shared feeding areas**

- Multiple sockeye salmon populations feed in the Pacific at the same time but remain distinct due to separate breeding/migratory systems; disturbances can suddenly alter connectivity and survival during migration.

8. Population ecology II

A. Patterns of population growth

1. **Exponential** (J-shaped): growth rate $\propto$ N; occurs temporarily when conditions are favorable (e.g., good years, new colonization).
    a. Classic colonizer example: cattle egret—after arrival in new regions, numbers rose exponentially for years.
2. **Logistic** (S-shaped): rapid initial increase, then levels off near carrying capacity K;
    a. real populations often only roughly match an S. Tasmanian sheep show the rise then hover/oscillate around a maximum.



3. **Fluctuations** (most common): sizes rise/fall around a mean or around an expected trend (even during otherwise exponential/logistic phases).
    a. Lake Erie phytoplankton can spike or crash within days;
    b. outbreaks (e.g., mountain pine beetle) can be orders of magnitude larger than lows.
4. **Cycles**: regular high–low oscillations
    a. (e.g., lemmings/voles ~3–5-yr cycles). Drivers may be predators, food, weather, or combinations; mechanisms vary by place/species.
    b. Cycles can weaken or collapse with environmental change (e.g., altered snow structure, warming), or from specialist pathogens.

B. Exponential vs. Logistic growth

1. **Exponential (continuous):**

$$\frac{dN}{dt} = rN, \qquad N(t) = N_0 e^{rt}$$

- Applies with continuous reproduction and effectively unlimited resources for a period

2. **Logistic (density‑dependent)**

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$$

- r: intrinsic rate under ideal conditions; K: carrying capacity (environment's long-term supportable size).
- When N≪K, logistic ≈ exponential (because $1-N/K \approx 1$)
- As N→K, $1-N/K \to 0$, so growth slows to zero
- Maximum growth rate occurs at N=K/2
- Closed-form logistic solution (predict N at time t):

$$N(t) = \frac{K}{1 + \left(\frac{K - N_0}{N_0}\right) e^{-rt}}$$

C. Density‑independent vs. density‑dependent factors

- Density‑independent: effects on births/deaths don't depend on N.
    - Examples: temperature, rainfall, storms, fires;
    - e.g., thrips abundance in Australia predicted from weather alone in some years; multi-decadal tree mortality trends associated with warming/"hotter droughts".
- Density‑dependent: births/deaths/dispersal change with N (competition for food/space, disease, predation).

- Regulation: factors increase growth when N is low and decrease it when N is high; only density-dependent factors regulate populations.
- Examples:
    - Song sparrows (Mandarte I.): fewer young/female at higher density; food addition boosted young produced (causal test).
    - Soybeans: high planting density → self-thinning (many die).
    - Flour beetles (Tribolium): mortality increases with egg density.
    - Poa annua (λ) and Daphnia (r): growth rate declines with density (diagnosis of density dependence).

D. Case studies & consequences

- Outbreak dynamics: Black Sea comb jelly boom via prey increase; mountain pine beetle killed forests over tens of millions of hectares, altering carbon balance.
- Disease & fluctuations: 1993 Hantavirus outbreak linked to rain → plant growth → mouse population boom → human exposure.
- Conservation signal: Elephants—continent-scale growth rates dropped below $\lambda=1$ after 2009, largely due to poaching; reversing decline requires $\lambda>1$ (reduce illegal killing/demand).

E. Quick practice

a. Identify patterns
    - Cattle egret in a new region (1990s SF Bay): Exponential phase.
    - Tasmanian sheep over a century: Rough logistic with fluctuations.
    - Lemming 4-yr highs/lows: Cycles.

b. Density dependence checks
    - If food is limiting at high N, what happens to r (or λ)?
        - → Declines with increasing N; may cross $r=0$ (or $\lambda=1$) at equilibrium.

- Which factors regulate? → Density-dependent (not weather shocks alone).
- Regulation test: does r (or λ) decrease with N and cross 0 (or 1) at equilibrium? If yes → density-dependent regulation.

9. Population ecology III

Z. Key Concept

- The risk of extinction increases in populations that fluctuate in size and/or are small

A. What is delayed density dependence?

- **Idea**: Birth/survival at time t depends on past density (e.g., food made earlier, maturation time, predator response), not just current N.
- **Mechanism**: There's a lag (development, gestation, maturation, numerical predator response). So the population "over- or undershoots" what current resources can support, producing oscillations.
- **Examples** of delays:
  - predator numbers track prey with a lag;
  - parasites take time to impact hosts;
  - juveniles produced after a food pulse only join the breeders later.

B. Case Study: Nicholson's blowfly experiments (classic DDD)

a. Setup A (cycles appear): Adults had unlimited food; maggots limited (50 g/day).

- Many adults ⇒ enormous egg laying ⇒ when larvae hatch, severe food shortage ⇒ larval die-off ⇒ next adult cohort collapses.
- After the crash, few adults lay few eggs ⇒ enough food per larva ⇒ adult numbers rebound.
- Result: repeated rise–crash cycles driven by the lag from adult egg laying to larval crowding/mortality.

b. Setup B (cycles damped): Limit food for adults and larvae.

- Adults can't massively over-produce eggs ⇒ larval crowding less extreme ⇒ population fluctuates around a mean (~4,000), not boom-bust.

c. Take-home: When the negative feedback (resource shortage) is delayed to a later life stage, cycles emerge; remove or reduce the delay and cycles dampen.

C. Predator–prey intuition with lags

1. Low predators → prey increase.
2. With a lag, predators increase later, overshooting prey.
3. High predators depress prey → predator reproduction/survival then falls later.
4. Prey recovers, and the sequence repeats.
5. Internal (hormonal/behavioral) or external (weather/food/predators) drivers can contribute; mechanisms differ by system.

D. Why it matters: fluctuations → extinction risk

- All populations fluctuate; DDD can amplify those swings.
- Small N + high variance in λ (or r) ⇒ much higher chance of strings of bad years ⇒ extinction risk rises.
- Lecture emphasis: larger environmental variance (e.g., SD of λ) sharply increases the probability of extinction; small populations are especially vulnerable.

E. How to recognize DDD (practical cues)

- Timing: Peaks in births or juvenile recruitment follow earlier high densities (or resource pulses) by a fixed developmental delay.
- Phase lag: In time series, the driver (e.g., food/prey) peaks before the response (consumers/predators).
- Experimental cue: If you cap the overproduction stage (e.g., limit adults) or buffer juveniles, cycles dampen.

F. Birth & Death Rate

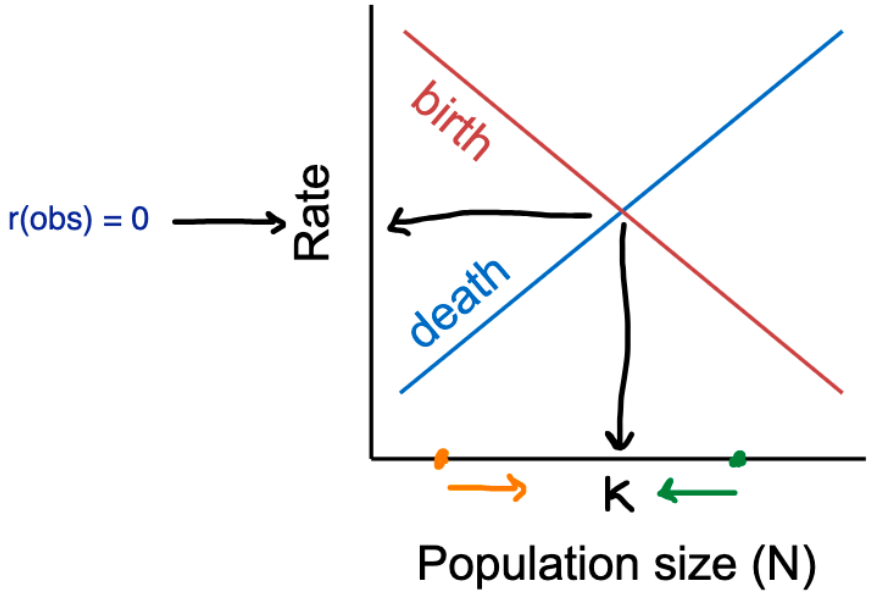
- N – current population size (number of individuals). Used in growth formulas and on the x-axis of density-dependence graphs.
- r_{max} – the per-capita intrinsic rate of increase under ideal conditions (i.e., when the population is uncrowded).
- r_{obs} – the per-capita intrinsic rate of increase under actual conditions (the realized growth rate the population is experiencing now).
- b, d – per-capita birth and death rates (events per individual per unit time). Slides define b and d and then relate them to r_{obs}
- K – the carrying capacity, the maximum population size the environment can support in the long run; at K, births and deaths balance.

$$N_{t+1} = N_t + B - D$$

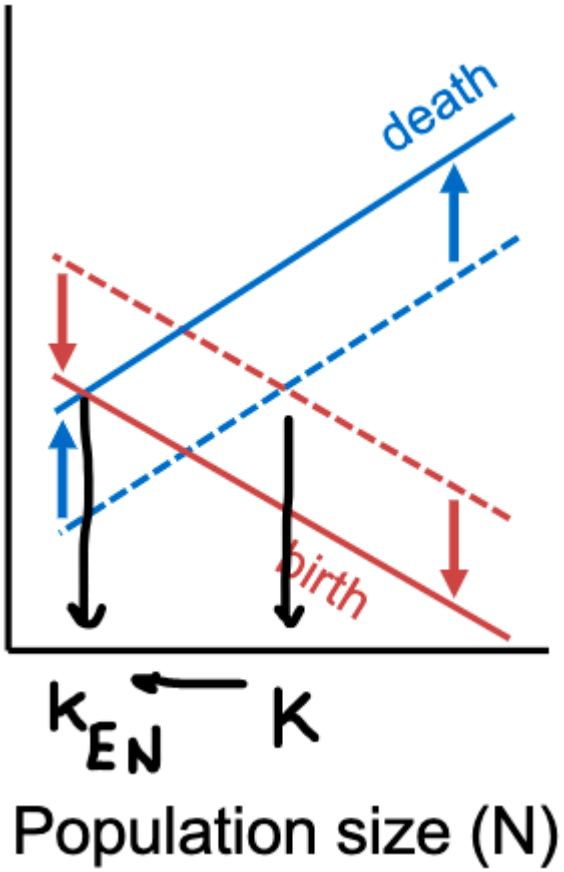
$$r_{obs} = b - d$$

At K, B = D (assuming no immigration or emigration).

Where is K on this graph?

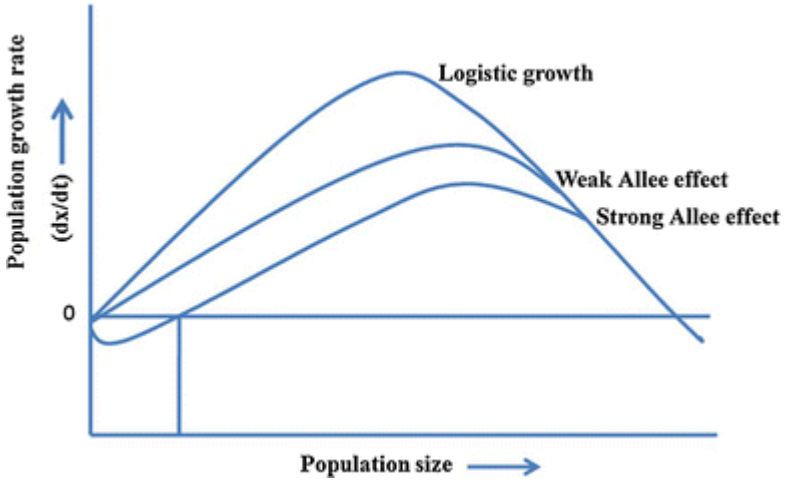


- Case Studies: El Niño causes a significant decrease in plankton populations: how do birth rates and death rates change during an El Niño year in graph?



G. Allee effects

- Definition: At low density, per-capita growth increases with N (small populations do worse).
- Why: mate-finding limits, loss of group protection/foraging, weak pollination.
- Slide cases: Bluefin tuna (schooling breakdown), kākāpō (mate limitation).
- **Implication:** Below an Allee threshold N_A, $r_{obs}<0$ → extinction risk ↑; management aims to push N **above** N_A



10. Review for Midterm 1

Text: The B